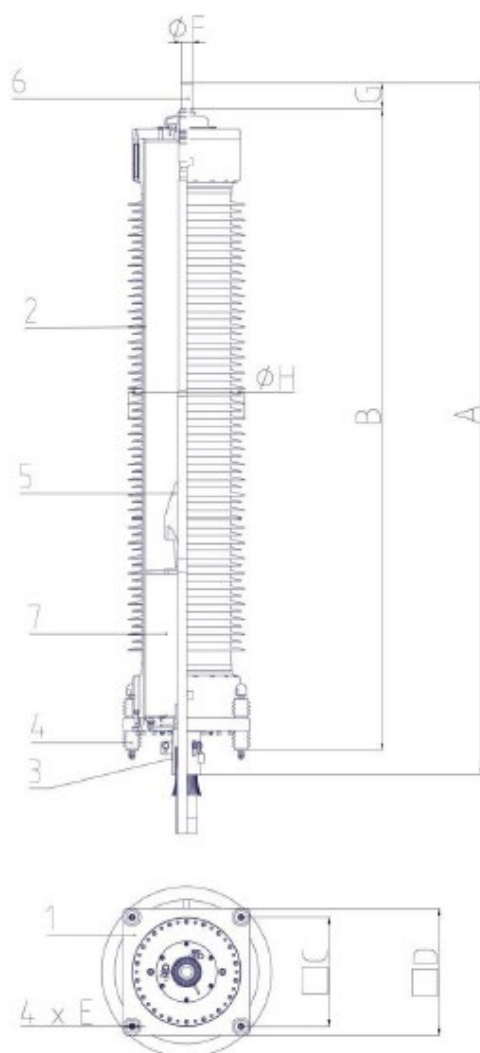


Project Description

Cable terminations are essential to connect cables to overhead lines, transformers etc. and typically are constructed in two ways, using geometrical field grading or capacitive field grading. In this project work, we will work on a termination with geometrical field grading (5 – stress relief cone in the figure below) that also uses compressed gas as insulation.



Main parts

Item	Name of part	Material
1	Base plate	Aluminium
2	Composite Insulator	Epoxy / Silicone
3	Introduction gland	Aluminium / Copper
4	Stand-off insulator	Glass composite
5	Stress relief cone	Elastomer
6	Connecting bolt	Aluminium / Copper
7	Insulating medium	Silicon fluid/gaz SF ₆

Fig. 1 – 245 kV Cable termination from Nexans (FRX 1.245 – 8300)

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Tasks

- Perform a literature review on the assigned gas (mixture):
 - Breakdown criteria for streamers in the gas itself and along insulating surfaces.
 - What is the gas pressure dependency?
 - For the outside of the composite insulator (2 in Fig. 1), what is the breakdown criteria for air. Assume dry conditions.
 - Stress relief cone. This one is subjected to secrets, and we do not have the exact design. Thus, you need to do a review on this kind of cones. The size in the drawing of the provided termination is approximately correct. Use this as a first iteration.
- What voltage levels is your termination subjected to during operation (LI, SI and temp. overvoltage). This can be found in relevant textbooks and IEC standards.
- Make a COMSOL model and MatLab/Phyton scripts to evaluate your solution (latter as appendix to the report). You should optimize your solution for the assigned insulation gas with a pressure that fulfils the criteria of no breakdown during the applied stresses (AC/LI/SI)
- Write and scientific report with your work.

The report have to be uploaded to Inspera using your candidate numbers on the front page. More information on this will come later.

Table 1 – Material properties

Material	Relative permittivity ϵ_r
EPDM	2.36
Fiberglass	4.4
SiR	4.3

PS: since this is the first time we're having a project counting towards the final grade, we will have to adjust a bit underway.